Four limit cycles in a planar three-reaction mass-action system: a proof of the Boros–Hofbauer conjecture

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Abstract. Boros and Hofbauer conjectured that the planar mass-action system of three reactions $\dot{x} = 1/y - x^a y^b$, $\dot{y} = K(-(d-1) - 1/y + dx^a y^b)$ has, for suitable parameters, an asymptotically stable equilibrium surrounded by four limit cycles, after locating numerically a point where the first three focal values vanish and the fourth is negative; they could not handle L_2, L_3, L_4 analytically. We prove the conjecture by two independent computer-assisted arguments. First, we compute the focal values exactly as polynomials over $\mathbb{Q}$ in a square-root-free parametrisation, certify with a Krawczyk operator in ball arithmetic a simple common zero of L_1, L_2, L_3 with $L_4 \in [-2.58012649949 \cdot 10^{-6} \pm 10^{-34}]$, and conclude by successive perturbations. Second, for explicit decimal parameters, validated Taylor integration of the first-return map brackets four sign changes of the displacement function, giving four nested limit cycles with alternating stability. We also show that this order-four weak focus is the only point of the parameter domain where $L_1 = L_2 = L_3 = 0$.

1 The problem

B. Boros, J. Hofbauer, *Oscillations in planar deficiency-one mass-action systems*, J. Dyn. Diff. Equat. 36 (Suppl 1):175–197 (2024), doi:10.1007/s10884-021-10051-z, arXiv:2103.00972 (also PMC10901957), §5.1 ("Four limit cycles"). They consider the three-reaction network (15) with $(a_1, b_1) = (0, 0)$, $(a_2, b_2) = (0, -1)$, $(a_3, b_3) = (a, b)$, $(c_1, d_1) = (0, -1)$, $(c_2, d_2) = (1, -1)$, $(c_3, d_3) = (-1, d)$, $a > 0$, $b > -1$, $d > 0$, $1 + bd > 0$. After scaling the equilibrium to $(1, 1)$ (possible iff $d > 1$) the mass-action ODE (16) becomes

$$\dot{x} = \frac{1}{y} - x^a y^b, \qquad \dot{y} = K\Big(-(d-1) - \frac{1}{y} + d\, x^a y^b \Big), \qquad K > 0. \tag{BH}$$

Verbatim (LaTeX source of arXiv:2103.00972, `sections/5_three_reactions.tex`, §5.1; the same text is in the published version):

> "In this subsection we discuss why we strongly conjecture that there exist parameter values for which the differential equation (16) has at least 4 limit cycles. [...] Set $K = \frac{a}{1+bd}$ to make $\operatorname{tr} J$ equal to zero. The first focal value, L_1, is then $L_1 = \frac{\pi a(2a^2 d + a(1+bd) - (1+bd)^2)}{8a\sqrt{(d-1)(1+bd)^3}}$ [...] Thus, there exist a, b, d such that $L_1 = L_2 = L_3 = 0$ (namely, we numerically find that this happens at $a = \widehat{a} \approx 1.01282$, $b = \widehat{b} \approx 0.65463$, $d = \widehat{d} \approx 3.28862$). Since, again numerically, we see that L_4 is negative for $\widehat{a}$, $\widehat{b}$, $\widehat{d}$, we conjecture that there exist parameter values for which

> the unique positive equilibrium of the differential equation (16) is asymptotically stable, and is surrounded by four limit cycles Γ_0, Γ_1, Γ_2, Γ_3, which are unstable, stable, unstable, stable, respectively. Since the formulas for L_2, L_3, and L_4 get complicated, we cannot handle them analytically. This is why we leave the existence of four limit cycles a conjecture."

2 Results

Parameter domain: $\mathcal{P} = \{p = (K, a, b, d) :\ K > 0,\ a > 0,\ b > -1,\ d > 1,\ 1 + bd > 0\}$ (open). For $p \in \mathcal{P}$, $(1, 1)$ is the unique positive equilibrium of (BH), with Jacobian

$$J(p) = \begin{pmatrix} -a & -(1+b) \\ Kda & K(1+bd) \end{pmatrix}, \qquad \det J = Ka(d-1) > 0, \qquad \operatorname{tr} J = -a + K(1+bd).$$

L_k denotes the k-th focal value (Lyapunov quantity) at $(1, 1)$ on the hypersurface $\mathcal{T}_0 = \{\operatorname{tr} J = 0\} = \{K = a/(1 + bd)\}$, in the normalisation of Boros–Hofbauer (§4.4 below shows that our L_k coincide with theirs on all formulas and values they give).

Theorem A (four limit cycles). There is a parameter point $p^* = (K^*, a^*, b^*, d^*) \in \mathcal{T}_0$ with

$$a^* = 1.012821998680314600068208268899 0\ldots, \quad b^* = 0.654625621104238210727259033435 9\ldots,$$
$$d^* = 3.288610439309935787724161356232 6\ldots, \quad K^* = a^*/(1 + b^*d^*) = 0.321244360398960809606413563359$$

(truncations of Arb enclosures whose radii are $\le 2{\cdot}10^{-39}$, see `certificates/certificate_output.json`) such that

1. $L_1(p^*) = L_2(p^*) = L_3(p^*) = 0$;
2. $L_4(p^*) \in [-2.58012649949442448635653476 27 \cdot 10^{-6} \pm 10^{-34}]$, in particular $L_4(p^*) < 0$;
3. the Jacobian matrix $\partial(L_1, L_2, L_3)/\partial(a, b, d)$ at (a^*, b^*, d^*) is nonsingular.

Consequently, every neighbourhood of p^* in $\mathcal{P}$ contains parameters p for which the equilibrium $(1, 1)$ of (BH) is a hyperbolic, asymptotically stable focus surrounded by (at least) four limit cycles $\Gamma_0, \Gamma_1, \Gamma_2, \Gamma_3$ (nested, Γ_0 innermost), where Γ_0 and Γ_2 are repelling (unstable) and Γ_1, Γ_3 are attracting (stable). The four cycles can be taken inside any prescribed neighbourhood of $(1, 1)$.

Corollary (the conjecture of Boros–Hofbauer). For the network of §5.1 with parameters (K, a, b, d) as produced by Theorem A (or the explicit $p^\sharp$ of Theorem C below) and rate constants $\kappa_1 = d - 1$, $\kappa_2 = 1$, $\kappa_3 = K^{-a}$, the mass-action system (16),

$$\dot{x} = \kappa_2 y^{-1} - \kappa_3 x^a y^b, \qquad \dot{y} = -\kappa_1 - \kappa_2 y^{-1} + d\,\kappa_3 x^a y^b,$$

has the unique positive equilibrium $(K, 1)$, which is asymptotically stable and surrounded by four limit cycles that are unstable, stable, unstable, stable (from the inside out). Indeed, $x = KX$, $y = Y$, $t = K\tau$ transforms it into (BH) (a linear change of coordinates and a constant positive time rescaling). Hence the conjecture of §5.1 holds, in the precise form stated there.

Theorem B (uniqueness; bonus). In the whole domain $\mathcal{P}$, the set $\{\operatorname{tr} J = 0,\ L_1 = L_2 = L_3 = 0\}$ is the single point p^*. Consequently, for every $p \in \mathcal{T}_0$ the equilibrium $(1,1)$ is a weak focus of order at most 4 (order exactly 4 only at p^*); in particular it is never a centre in this family.

Theorem C (explicit parameters; second, independent proof). Let $p^\sharp = (K, a, b, d)$ be the exact decimal fractions

$$\begin{aligned} a &= 1.012822827566885038677856847030028159177026555316278466298 37, \\ b &= 0.654626809470286327709996401918672391083172153706469077547131, \\ d &= 3.288605466176904297183887415659965416086564877981034478548 74, \\ K &= 0.321244556816175423011532938999913745383501015965226302231941 \end{aligned}$$

(`certificates/explicit_params.json`; $\operatorname{tr} J = -1.90 \cdot 10^{-22} < 0$). Then the equilibrium $(1,1)$ of (BH) is a hyperbolic stable focus and the system has (at least) four limit cycles $\Gamma_0, \dots, \Gamma_3$, crossing $\Sigma = \{(1+\rho, 1) : \rho > 0\}$ at $\rho \in (0.002, 0.004)$, $(0.004, 0.006)$, $(0.006, 0.012)$, $(0.012, 0.024)$ respectively, Γ_0, Γ_2 repelling and Γ_1, Γ_3 attracting. The proof (§6.4) uses only validated (interval) Taylor integration of the ODE and planar topology; it does not use the focal values.

Remark (not used, not re-proved). With the classical finite-cyclicity theory of a weak focus (Bautin 1952; see e.g. Roussarie 1998, Han–Yu 2012), Theorem B implies that at most four small-amplitude limit cycles can bifurcate from $(1,1)$ in this family, and that for parameters close enough to p^* as in the proof of Theorem A there are exactly four such cycles, all hyperbolic. Our proofs of Theorems A and B do not depend on this remark.

What is proved by what:

claim	method	status
exact polynomials G_k $(k \le 4)$ defining L_k	exact rational arithmetic (FLINT)	proved, modulo correctness of a ~250-line program (validated in §4.4, §8)
$L_1 = L_2 = L_3 = 0$ at a unique point of a box, Jacobian nonsingular	Krawczyk operator in Arb ball arithmetic	proved (computer-assisted)
$L_4 < 0$ there	Arb enclosure	proved (computer-assisted)
sign lemma, return map, successive perturbations	pen-and-paper (§4.2, §6)	proved
Theorem B	exact resultants + Arb root enclosures	proved (computer-assisted)
Theorem C (explicit rational parameters, four cycles bracketed on Σ)	validated Taylor integration in Arb + Jordan-curve/Poincaré–Bendixson argument	proved (computer-assisted), independent of the focal-value code
location of the cycles to more digits ($\rho \approx 0.004, 0.008, 0.016, 0.032$ on another ray)	60-digit floating-point Taylor integration	numerical illustration only

3 Set-up and reduction

3.1 Sqrt-free parametrisation of $\mathcal{T}_0$. On $\mathcal{T}_0$ the eigenvalues of J are $\pm i\omega$ with $\omega^2 = \det J = a^2(d-1)/(1+bd)$. Put

$$c := 1 + b > 0, \qquad s := \frac{\omega^2}{a^2} = \frac{d-1}{1+bd} > 0, \qquad w := \sqrt{s}, \qquad \rho := 1 - (c-1)s.$$

Then $\rho = (1+b)/(1+bd) > 0$, $d = (1+s)/\rho$, $1 + bd = c/\rho$, $K = a\rho/c$, $K(d-1) = as$, $Kd = a(1+s)/c$. The map $\Psi : (a, c, s) \mapsto (K, a, b, d) = (a\rho/c,\ a,\ c-1,\ (1+s)/\rho)$ is a real-analytic bijection from $\mathcal{D} = \{a > 0,\ c > 0,\ s > 0,\ \rho > 0\}$ onto $\mathcal{T}_0$; the map $(a, b, d) \mapsto (a, c, s)$ has Jacobian determinant $\partial s/\partial d = (1+b)/(1+bd)^2 > 0$.

3.2 Normal coordinates. Let $u = x - 1$, $v = y - 1$, $I(v) = (1+v)^{-1}$, $E(u, v) = (1+u)^a(1+v)^b$, $f = I - E$. Using $\dot{y} = K[(d-1)(I-1) - df]$, the linear change and time rescaling

$$u = cX, \qquad v = a(wY - X), \qquad \tau = \omega t = aw\, t$$

transform (BH) (for $p = \Psi(a, c, s)$) into

$$X' = -Y + P(X, Y), \qquad Y' = X + Q(X, Y), \qquad P = \frac{N_f}{acw}, \quad Q = \frac{cN_I - N_f}{ac}, \tag{N}$$

where N_I, N_f are the parts of degree ≥ 2 in (X, Y) of I and f (direct check: the linear parts are $-Y$ and X). Every Taylor coefficient of N_I, N_f is a polynomial in (a, c, w) because $(1+u)^a(1+v)^{c-1} = \sum_{i,j} \binom{a}{i}\binom{c-1}{j} u^i v^j$. The series converge for $|u|, |v| < 1$, so (N) is real-analytic near the origin, jointly in the parameters.

4 Focal values

4.1 Algorithm (complex Lyapunov-function method). Write $z = X + iY$; (N) reads $z' = iz + R(z, \bar{z})$, $R = P + iQ = \sum_{p+q\geq 2} R_{pq} z^p \bar{z}^q$. We look for a real polynomial $F = z\bar{z} + \sum_{3\leq p+q\leq N} f_{pq} z^p \bar{z}^q$ ($f_{qp} = \overline{f_{pq}}$). Along (N),

$$\dot{F} = \sum i(p-q) f_{pq} z^p \bar{z}^q + \sum f_{rs}\big(r z^{r-1}\bar{z}^s R + s z^r \bar{z}^{s-1}\bar{R}\big).$$

Let S_{pq} be the coefficient of $z^p\bar{z}^q$ in the second sum computed with the f of degree $< p+q$ (only those contribute, because $R = O(|z|^2)$). Degree by degree, $n = 3, 4, \ldots$:

$$f_{pq} = \frac{i\, S_{pq}}{p-q}\ \ (p \neq q), \qquad V_k := S_{k+1,k+1} \in \mathbb{R}\ \ (n = 2k+2), \qquad f_{k+1,k+1} := 0.$$

Then, for every $j \geq 1$, the polynomial F_j built up to degree $2j+2$ satisfies

$$\dot{F}_j = \sum_{k=1}^{j} V_k\, |z|^{2k+2} + O(|z|^{2j+3}). \tag{4.1}$$

(The terms of degree $\le 2j+2$ of $\dot F_j$ are exactly the ones cancelled or recorded by the recursion; all remaining terms have degree $\ge 2j+3$ because F_j is a polynomial and $R = O(|z|^2)$ is analytic.) The choice $f_{k+1,k+1} = 0$ is a convention; Lemma 4.2 holds for any choice.

4.2 Sign lemma. *Let $p \in \mathcal{T}_0$ and suppose $V_1(p) = \cdots = V_{j-1}(p) = 0 \neq V_j(p)$. Let Σ be any ray issuing from $(1,1)$ and transversal to the flow near $(1,1)$ (e.g. $\{(1+\rho,1) : \rho > 0\}$), and $\delta(\rho) = \mathcal{P}(\rho) - \rho$ the displacement function of the first-return map on Σ (§6.1). Then $\operatorname{sign}\delta(\rho) = \operatorname{sign} V_j(p)$ for all sufficiently small $\rho > 0$.*

Proof. By (4.1), $\dot F_j = V_j|z|^{2j+2}(1 + O(|z|))$, so $\dot F_j$ has the sign of V_j on a punctured disc $0 < |z| < r_0$. Since $F_j = |z|^2 + O(|z|^3)$, F_j is positive definite and its restriction to the ray Σ is strictly increasing in ρ for small ρ ($F_j = q\rho^2 + O(\rho^3)$, $q > 0$, in any linear coordinates). For ρ small, the orbit from ρ stays in the punctured disc during one turn (§6.1), so $F_j(\mathcal{P}(\rho)) - F_j(\rho) = \int \dot F_j\, dt$ has the sign of V_j, hence so does $\mathcal{P}(\rho) - \rho$. □

(When $V_1 = \cdots = V_{j-1} = 0$, (4.1) gives $d(|z|^2)/d\tau \approx V_j|z|^{2j+2}$, i.e. $\delta(r) = \pi V_j r^{2j+1} + O(r^{2j+2})$ in the coordinate $r = |z|$ of (N). Using the unscaled coordinate $X = u$ instead of $u = cX$ gives the normalisation of the paper, $L_j = \pi V_j^{(u)}$; see §4.3.)

4.3 Exact polynomials. Put $D = acw$ and $\tilde R := D\,R = N_f + iw(cN_I - N_f)$ (polynomial coefficients). Running the recursion on $\tilde f_{pq} := D^{\,p+q-2} f_{pq}$ gives the polynomial recursion

$$\tilde S_n = \sum_{m=2}^{n-1} D^{\,n-1-m}\big[\tilde F_m \otimes \tilde R_{n-m+1}\big], \qquad \tilde f_{pq} = \frac{i\tilde S_{pq}}{p-q}, \qquad \tilde V_k = \tilde S_{k+1,k+1} = D^{2k} V_k,$$

so that $\tilde V_k \in \mathbb{Q}[a,c,w]$. The program `scripts/focal_exact.py` (python-flint `fmpq_mpoly`, complex numbers as pairs of polynomials) computes $\tilde V_1, \dots, \tilde V_4$ in 0.1 s and checks along the way that every $\tilde S_{k+1,k+1}$ is real and that $\tilde S_{qp} = \overline{\tilde S_{pq}}$. Every $\tilde V_k$ is w times a polynomial in (a,c,w^2); we set

$$G_k(a,c,s) := \tilde V_k(a,c,\sqrt{s})/\sqrt{s} \in \mathbb{Q}[a,c,s], \qquad L_k = \pi\, V_k^{(u)} = \frac{\pi\, G_k(a,c,s)}{a^{2k} c^{4k}\, s^{\,k-1/2}},$$

where $V_k^{(u)}$ is the Lyapunov quantity in the coordinates $X = u$ (unscaled), i.e. $V_k^{(u)} = c^{-2k} V_k$. On $\mathcal{D}$, $\operatorname{sign} L_k = \operatorname{sign} V_k = \operatorname{sign} G_k$, so Lemma 4.2 holds with G_k in place of V_k. Factorisations (exact):

$$G_1 = -\tfrac{1}{8}\, a^2 c^2\, g_1, \quad G_2 = \tfrac{1}{1152}\, a^4 c^4\, g_2, \quad G_3 = -\tfrac{1}{442368}\, a^6 c^6\, g_3, \quad G_4 = \tfrac{1}{530841600}\, a^8 c^8\, g_4,$$

with $g_k \in \mathbb{Z}[a,c,s]$ irreducible, of degrees $(2k, 2k, 3k-1)$ in (a,c,s) and 9, 48, 136, 296 terms. Explicitly

$$-g_1 = 2a^2(1+s)\rho + ac\rho - c^2, \qquad \rho = 1 + s - cs,$$

and g_2, g_3 are listed in Appendix A; all four are stored in `certificates/G_k.json`.

4.4 Validation against the paper. (i) Exact identity (`crosscheck.py`): $L_1 = \pi G_1/(a^2c^4\sqrt{s}) = \pi[2a^2(1+s)\rho + ac\rho - c^2]/(8c^2\sqrt{s})$ equals the paper's $\pi a(2a^2d + a(1+bd) - (1+bd)^2)/(8a\sqrt{(d-1)(1+bd)^3})$ after $d = (1+s)/\rho$, $1+bd = c/\rho$. (ii) The generic floating-point implementation `focal_mp.py` (independent code path: numerical Taylor expansion of arbitrary generalized-monomial systems, numerical normalisation with $\omega = \sqrt{\det J}$, no reparametrisation) reproduces **exactly** (ratio 1.000… to 15+ digits) every focal value printed in the paper: L_1 of §3.2, §4.1 and §5.1 (two generic points each), L_2 of §4.1 along $\{L_1 = 0\}$ (two points), $L_2 \approx 0.01293$ of §3.2, and $L_3 = -\frac{625\pi}{110592}\sqrt{7/2}$ of §4.1. (iii) The exact L_k agree with `focal_mp.py` at four parameter points to relative precision 10^{-55} for $k = 1..4$. (iv) The paper's checkpoints: at $(a, d) = (1, 165/49)$ (then $b = 7/11$, $s = 58/77$) we get exactly $L_1 = L_2 = 0$ and $L_3 < 0$ (exact rational sign); at $(a, d) = ((1+\sqrt{3961})/60, 3)$, $L_1 = L_2 = 0$ (to 50 digits) and $L_3 > 0$ — as stated in the paper.

5 The interval certificate (`scripts/certify.py`)

Krawczyk's theorem (Krawczyk 1969; Moore 1977; Neumaier 1990, §5.1). Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be C^1 on a box X, $m \in X$, C a real $n \times n$ matrix and $\mathbf{J}$ an interval matrix containing $G'(x)$ for all $x \in X$. If $\mathcal{K}(X) = m - CG(m) + (I - C\mathbf{J})(X - m) \subset \operatorname{int} X$, then G has exactly one zero in X. Moreover, if $\|I - C\mathbf{J}\|_\infty < 1$, every matrix in $\mathbf{J}$ is nonsingular (Neumann series for CA).

Data (`certificates/box.json`): variables (a, c, s), centre m = three rationals with denominator $\le 10^{60}$ (Newton iterate at 150 digits), radius $r = 10^{-40}$, preconditioner C = rational approximation of $G'(m)^{-1}$. Computations in Arb (python-flint 0.9.0, 512-bit midpoint–radius balls with outward rounding); $G = (G_1, G_2, G_3)$, $\mathbf{J}$ = exact derivative polynomials evaluated on X.

check	result
C0 recomputed G_k = stored G_k; C0′ Arb balls contain the exact box	PASS
C1 on X: $a > 0$, $c > 0$, $s > 0$, $\rho > 0$, $d > 1$, $1 + bd > 0$, $K > 0$	PASS
C2 $\mathcal{K}(X) \subset \operatorname{int} X$: $\max_i \lvert\mathcal{K}_i - m_i\rvert \le 3.62 \cdot 10^{-61} \ll r = 10^{-40}$	PASS
C3 $\|I - C\mathbf{J}\|_\infty \le 1.37 \cdot 10^{-32} < 1$; $\det \mathbf{J} \in [-0.208483123515217 \pm 4 \cdot 10^{-16}]$	PASS
C4 $G_4(X) \subset [-9.3543144375364141858217679 70 \cdot 10^{-4} \pm 2 \cdot 10^{-32}] < 0$; $L_4(X) \subset [-2.5801264994944244863565347627 \cdot 10^{-6} \pm 10^{-34}]$	PASS
C5 $\det \partial(G_1, G_2, G_3)/\partial(a, b, d) \in [-0.03470367143 \pm 4 \cdot 10^{-12}]$	PASS

Let $\mu^* \in X$ be the unique zero and $p^* = \Psi(\mu^*)$. At μ^*, $G_1 = G_2 = G_3 = 0$, so $L_1 = L_2 = L_3 = 0$; $L_4(p^*) < 0$ by C4; and since $L_k = \pi G_k/(\text{positive})$, at a common zero $\partial L_k = \pi\, \partial G_k/(\text{positive})$, so C5 gives item 3 of Theorem A. The enclosures of (a^*, b^*, d^*, K^*) quoted in §2 are the Arb images of X (`certificates/certificate_output.json`). The

paper's values $a \approx 1.01282$, $b \approx 0.65463$, $d \approx 3.28862$ agree with ours except in the last digit of d ($d^* = 3.288610\ldots$), which reflects their numerical accuracy.

6 The bifurcation argument (proof of Theorem A) and validated integration (proof of Theorem C)

Let $\Sigma = \{(1+\rho, 1) : \rho > 0\}$. On Σ, $\dot{y} = Kd((1+\rho)^a - 1) > 0$: Σ is transversal.

6.1 Lemma (return map). Let $p_0 \in \mathcal{T}_0$. There are a compact neighbourhood $U \subset \mathcal{P}$ of p_0 and $\bar{\rho} > 0$ such that for all $p \in U$ and $\rho \in (0, \bar{\rho}]$ the forward orbit of $(1+\rho, 1)$ returns to Σ after one turn around $(1,1)$, at $(1 + \mathcal{P}(\rho, p), 1)$; the map $\mathcal{P}$ is continuous on $(0, \bar{\rho}] \times U$, real-analytic in ρ, and strictly increasing in ρ. Each zero $\rho_0 \in (0, \bar{\rho}]$ of $\delta(\cdot, p) = \mathcal{P}(\cdot, p) - \mathrm{id}$ gives a periodic orbit through $(1+\rho_0, 1)$ surrounding $(1,1)$; distinct zeros give distinct periodic orbits, and an isolated zero gives an isolated periodic orbit, i.e. a limit cycle.

Proof (standard). Let T bring $J(p_0)$ to the rotation form and use polar coordinates (r, θ) for $\xi = T(u, v)$. Since $J(p) \to J(p_0)$ as $p \to p_0$ and the nonlinear terms are $O(r^2)$ uniformly on a compact parameter set, there are U, $r_0 > 0$ with $\dot{\theta} \geq \omega_0/2$ and $|\dot{r}| \leq Lr$ for $r \leq r_0$, $p \in U$. For $r(0) \leq \bar{r} := r_0 e^{-4\pi L/\omega_0}$ the orbit stays in $r \leq r_0$ up to time $4\pi/\omega_0$, during which θ increases by at least 2π; hence it meets the ray Σ (a ray in the ξ-plane too) again. The crossings are transversal, so the implicit function theorem gives continuity in (ρ, p) and analyticity in ρ. Monotonicity follows from uniqueness of solutions (orbits do not cross). A periodic orbit in $\{r \leq r_0\}$ has winding number ± 1 around $(1,1)$ and θ is strictly increasing along it, so it meets Σ exactly once per period; a periodic orbit near one through ρ_0 meets Σ near ρ_0, hence is the same orbit if ρ_0 is an isolated zero. □

6.2 Lemma (trace). If $p \in U$ and $\operatorname{tr} J(p) < 0$, then $\delta(\rho, p) < 0$ for all small $\rho > 0$.
Proof. $(1,1)$ is a hyperbolic sink ($\det J > 0$), attracting a neighbourhood. If $\delta(\rho, p) = 0$ there would be a periodic orbit in that neighbourhood; if $\delta(\rho, p) > 0$, the iterates $\mathcal{P}^n(\rho)$ would increase (monotonicity) and not tend to 0. Both contradict attraction. □

6.3 Proof of Theorem A. Let $G = (G_1, G_2, G_3)$. By C2–C3, $G(\mu^*) = 0$ and $G'(\mu^*)$ is invertible, so (inverse function theorem) G maps a neighbourhood N of μ^* diffeomorphically onto a neighbourhood W of 0; write $\Phi = \Psi \circ (G|_N)^{-1} : W \to \mathcal{T}_0$, continuous with $\Phi(0) = p^*$. Fix $U, \bar{\rho}$ from Lemma 6.1 for $p_0 = p^*$ and shrink W so that $\Phi(W) \subset U$. Given any neighbourhood $\mathcal{V}$ of p^* and $\varepsilon > 0$, we construct $p_4 \in \mathcal{V}$ and $0 < \rho_5 < \rho_4 < \rho_3 < \rho_2 < \rho_1 < \varepsilon$ with

$$\delta(\rho_5, p_4) < 0 < \delta(\rho_4, p_4), \quad \delta(\rho_3, p_4) < 0 < \delta(\rho_2, p_4), \quad \delta(\rho_1, p_4) < 0.$$

- Step 0. At p^*: $G_1 = G_2 = G_3 = 0$, $G_4 < 0$. Lemma 4.2 ($j = 4$) gives $\rho_1 \in (0, \min(\bar{\rho}, \varepsilon))$ with $\delta(\rho_1, p^*) < 0$.
- Step 1. By continuity of $\delta(\rho_1, \cdot)$ choose $t_3 > 0$ so small that $p_1 := \Phi(0, 0, t_3) \in \mathcal{V}$ still has $\delta(\rho_1, p_1) < 0$. At p_1: $G_1 = G_2 = 0 < G_3$, so by Lemma 4.2 there is $\rho_2 < \rho_1$ with $\delta(\rho_2, p_1) > 0$.
- Step 2. Choose $t_2 > 0$ so small that $p_2 := \Phi(0, -t_2, t_3) \in \mathcal{V}$ keeps the signs at ρ_1, ρ_2. At p_2: $G_1 = 0 > G_2$: choose $\rho_3 < \rho_2$ with $\delta(\rho_3, p_2) < 0$.

- Step 3. Choose $t_1 > 0$ so small that $p_3 := \Phi(t_1, -t_2, t_3) \in \mathcal{V}$ keeps the signs at ρ_1, ρ_2, ρ_3. At p_3: $G_1 > 0$: choose $\rho_4 < \rho_3$ with $\delta(\rho_4, p_3) > 0$.
- Step 4. Let $p_4 = p_3 - (\eta, 0, 0, 0)$ (decrease K), $\eta > 0$ so small that $p_4 \in \mathcal{V} \cap U$ keeps the signs at $\rho_1, \dots, \rho_4$. Now $\operatorname{tr} J(p_4) = -\eta(1 + bd) < 0$; Lemma 6.2 gives $\rho_5 < \rho_4$ with $\delta(\rho_5, p_4) < 0$.

(All the $\Phi(\cdot)$ points lie on $\mathcal{T}_0$, where Lemma 4.2 applies; continuity of Φ makes each new point as close to the previous one as needed.) Since $\delta(\cdot, p_4)$ is analytic on $(0, \bar{\rho}]$ and not identically zero, it has finitely many zeros in $[\rho_5, \rho_1]$. On each of the four intervals $(\rho_5, \rho_4), (\rho_4, \rho_3), (\rho_3, \rho_2), (\rho_2, \rho_1)$ it changes sign; take the zeros $\zeta_0 < \zeta_1 < \zeta_2 < \zeta_3$, one in each interval, at which the sign changes from $-$ to $+$ (ζ_0, ζ_2) resp. from $+$ to $-$ (ζ_1, ζ_3) when ρ increases (they exist because the finitely many gaps between consecutive zeros carry constant signs that start with $-$, resp. $+$, and end with the opposite sign). Let Γ_i be the periodic orbit through $(1 + \zeta_i, 1)$ — a limit cycle by Lemma 6.1. If $\delta < 0$ just inside and $\delta > 0$ just outside ζ_0, orbits near Γ_0 move away from it on both sides ($\mathcal{P}$ is increasing and fixes ζ_0): Γ_0 is repelling; likewise Γ_2. At ζ_1, ζ_3 the signs are reversed and the iterates converge monotonically to ζ_i from both sides: Γ_1, Γ_3 are attracting. The equilibrium is a hyperbolic sink ($\operatorname{tr} J < 0 < \det J$). The Γ_i are nested Jordan curves around $(1, 1)$ (each meets Σ once, at increasing ζ_i) and lie in any prescribed neighbourhood of $(1, 1)$ if ε is small. □

6.4 Proof of Theorem C (`scripts/validated_cycles.py`).

Validated integrator. With $E = x^a y^b$, $I = 1/y$, $J = 1/x$, (BH) lifts to the polynomial system $x' = I - E$, $y' = K(-(d-1) - I + dE)$, $E' = E(aJx' + bIy')$, $I' = -I^2 y'$, $J' = -J^2 x'$; from consistent initial data its solution projects onto that of (BH). One step of length h from a ball enclosure Z_k of the state: (i) find a box B with $Z_k + [0, h]\, F(B) \subset \operatorname{int} B$ (then every solution starting in Z_k exists on $[0, h]$ and stays in B: if it left B at a first time $t_1 \le h$, then $z(t_1) \in z_0 + t_1 \operatorname{conv} F(B) \subset \operatorname{int} B$, a contradiction); (ii) Taylor coefficients $z_j(Z_k)$, $j < N$, by the standard recurrences in Arb; (iii) Lagrange remainder $z_N(B)\tau^N$ (the N-th Taylor coefficient is a polynomial function of the state, and the intermediate state lies in B). Hence $Z(t_k + \tau) \in \sum_{j<N} z_j(Z_k)\tau^j + z_N(B)\tau^N$ for $\tau \in [0, h]$. Parameters: $N = 36$, $h = 0.05$ (146 steps per turn), 320-bit balls. The enclosures of δ do not change (15 digits) with $(N, h) = (28, 0.03)$ or $(44, 0.08)$.

Crossing bookkeeping. Σ is transversal ($\dot{y} = Kd((1+\rho)^a - 1) > 0$). In every step whose box B meets the line $y = 1$, the script requires either ($x < 1$ and $\dot{y} < 0$ on B) or ($x > 1$ and $\dot{y} > 0$ on B), so y is strictly monotone there and crossings are counted exactly: for every test point the orbit leaves Σ upwards, crosses the left half-line $\{x < 1, y = 1\}$ exactly once (downwards), and the first return to Σ is enclosed by bisection of the crossing time on the Taylor model (2^{-140} relative width), giving $\mathcal{P}(\rho)$ to about 10^{-38}.

Certified values (`certificates/validated_cycles_output.json`):

ρ	$\delta(\rho) = \mathcal{P}(\rho) - \rho$
0.002	$[-4.60438730999 \cdot 10^{-25} \pm 6.2 \cdot 10^{-38}]$
0.004	$[+1.27356613600 \cdot 10^{-24} \pm 1.9 \cdot 10^{-36}]$
0.006	$[-3.54914581811 \cdot 10^{-24} \pm 4.5 \cdot 10^{-36}]$
0.012	$[+1.41126309912 \cdot 10^{-22} \pm 4.5 \cdot 10^{-34}]$

$$0.024 \quad [-3.02911287653 \cdot 10^{-20} \pm 4.8 \cdot 10^{-32}]$$

(radii as printed by Arb with 12 significant digits, i.e. including the display rounding).

(also $\operatorname{tr} J < 0 < \det J$, $(\operatorname{tr} J)^2 < 4 \det J$ in exact rational arithmetic).

Topology. Let $\rho_1 = 0.024$, $q_1 = (1 + \rho_1, 1)$, γ the orbit arc from q_1 to its first return $q_1' = (1 + \mathcal{P}(\rho_1), 1)$, $\mathcal{P}(\rho_1) < \rho_1$. Use polar coordinates (r, θ) around $(1, 1)$ in the plane $(u, v) = (x - 1, y - 1)$, so that $\Sigma = \{\theta = 0\}$. The script certifies $u\dot{v} - v\dot{u} > 0$ on every enclosure box along γ, i.e. θ is strictly increasing along γ; with the crossing bookkeeping (γ meets $\{\theta = \pi\}$ once and $\{\theta = 0\}$ only at its end points), θ increases by exactly 2π, so γ is a polar graph $r = R(\theta)$, $0 \le \theta \le 2\pi$, with $R(0) = \rho_1 > R(2\pi) = \mathcal{P}(\rho_1)$. Hence $C = \gamma \cup [q_1', q_1]$ is a Jordan curve bounding the star-shaped region $D = \{r \le R(\theta),\ 0 < \theta < 2\pi\} \cup \{\theta = 0,\ r \le \rho_1\}$, which contains $(1, 1)$ and the whole segment $\{(1 + \rho, 1) : 0 < \rho \le \rho_1\}$. A point of the open segment (q_1', q_1) has $R(2\pi^-) = \mathcal{P}(\rho_1) < r < \rho_1 = R(0^+)$: the flow, which crosses Σ with increasing θ, enters D there; orbits cannot cross γ. So D is positively invariant. For every $\rho \in (0, \rho_1]$ the orbit of $(1+\rho, 1)$ stays in D, where $(1, 1)$ is the only equilibrium (it is the unique positive equilibrium); by Poincaré–Bendixson its ω-limit set is $(1, 1)$ (a focus: the orbit spirals and crosses Σ) or a periodic orbit surrounding $(1, 1)$ (which meets Σ transversally, so the orbit crosses Σ near it); a homoclinic loop is impossible since $(1, 1)$ is a sink. Thus the first return map $\mathcal{P}$ is defined on $(0, \rho_1]$, and by transversality it is continuous and real-analytic there. By the table, δ changes sign on each of the four intervals; as in §6.3 (finitely many zeros, choice of sign-changing zeros, monotonicity of $\mathcal{P}$) we get four limit cycles with the stated stabilities. □

The parameters $p^\sharp$ were found with the focal values (targets (G_1, G_2, G_3) prescribed so that the truncated displacement has roots at $0.004, 0.008, 0.016, 0.032$ on the ray $Y = 0$, see §8.2), but the proof of Theorem C does not depend on how they were found.

Remarks. (a) Only the network parameters (a, b, d) and $K = \bar{x}/\bar{y}$ are varied, exactly as in the paper's §§4.1 and 5.1; the conjecture is about "parameter values" of (16), including the exponents. (By arXiv:2406.13451, rate constants alone cannot unfold a codimension > 1 bifurcation in a three-reaction network without conservation laws, so varying the network parameters is necessary.) (b) The perturbation directions are the paper's: first L_3, then L_2, L_1, finally the trace via K.

7 Proof of Theorem B (`scripts/uniqueness.py`)

On $\mathcal{D}$, $G_1 = 0 \iff 2a^2(1 + s)\rho + ac\rho - c^2 = 0$. With $t := a/c > 0$ this reads $\rho\, t(2t(1 + s) + 1) = 1$, i.e. $\rho = 1/(tQ)$, $Q = 2t(1 + s) + 1$, and $c = (1 + s - \rho)/s$. Hence $(t, s) \mapsto (a, c, s) = (tc, c, s)$ is a bijection from $\{t > 0,\ s > 0,\ 2t(1 + s) > 1\}$ (the last condition is $c > 0$) onto $\{G_1 = 0\} \cap \mathcal{D}$. Substituting and clearing the denominator $(stQ)^m$:

- G_1 becomes identically 0 (check);
- $H_2 = \text{const} \cdot (\text{powers of } s,\ s + 1,\ t,\ ts + t + 1,\ 2ts + 2t - 1) \cdot h_2(t, s)$, and the same for H_3 with h_3; all the listed small factors are nonzero on the (t, s)-domain ($2t(1 + s) - 1 > 0 \iff c > 0$); h_2, h_3 have degrees $(6, 4)$, $(10, 8)$ and 24, 68 terms.
- $R_t = \operatorname{Res}_s(h_2, h_3)$ (degree 58) factors as $t^{32}(t+1)^9(2t-1)^4(2t+3)(3t-1)\, p_{11}(t)$ with

$p_{11} = 628224t^{11} + 1764800t^{10} - 1694592t^9 - 15308848t^8 - 25081168t^7 + 10994160t^6 + 36659948t^5 - 2189226t^4 - 16683601t^3 + 3894133t^2 - 2248520t + 1436225$; $R_s = \mathrm{Res}_t(h_2, h_3)$ has degree 56.

- Any common zero (t_0, s_0) satisfies $R_t(t_0) = R_s(s_0) = 0$. All complex roots of both resultants are enclosed in Arb balls; the candidates with possibly positive real part and possibly zero imaginary part are $t \in \{1/3,\ 0.61212,\ 0.61956,\ 0.74045,\ 2.73230,\ 1/2\}$ and $s \in \{0.11924,\ 0.72590\}$. For all pairs but one, the Arb enclosure of h_2 or h_3 excludes 0. The surviving pair, mapped back to (a, c, s), lies inside the Krawczyk box X of §5, where the zero is unique.

Hence $\{G_1 = G_2 = G_3 = 0\} \cap \mathcal{D} = \{\mu^*\}$, which is Theorem B (the order statement follows from $L_4(p^*) \neq 0$). (PARI/GP gives the same picture: over $\mathbb{Q}(\theta)$, $p_{11}(\theta) = 0$, $\gcd(h_2, h_3)$ is linear in s; the real roots $\theta \approx 0.6196, 0.7405, 2.7323$ give $s < 0$.)

8 Non-rigorous cross-checks (not part of the proof)

1. **Independent method** (`polar_check.py`): the coefficients $u_k(2\pi)$ of the displacement map, obtained by integrating the triangular ODE system of the polar expansion $dr/d\theta = \sum C_k(\theta) r^k$ with DOP853 (tolerance 10^{-13}), agree with πV_k: $u_3 = \pi V_1$ at a generic point (rel. error $6 \cdot 10^{-15}$); on $\{L_1 = 0\}$: $u_3 \approx 0$, $u_5 = \pi V_2$ (10^{-11}); on $\{L_1 = L_2 = 0\}$: $u_5 \approx 0$, $u_7 = \pi V_3$ ($8 \cdot 10^{-11}$); at p^*: $u_2, \ldots, u_8 = O(10^{-14})$ and $u_9(2\pi) = -2.5801263 \cdot 10^{-6} = \pi V_4$ ($5 \cdot 10^{-8}$).
2. **Explicit four cycles** (`illustrate_cycles.py`, 60-digit arithmetic, Taylor integrator of order 40 on the polynomial lift $(x, y, x^a y^b, 1/y, 1/x)$): targets chosen so that the truncated displacement has roots $0.004, 0.008, 0.016, 0.032$; parameters $a = 1.01282282756688503867785684703002 8\ldots$, $b = 0.65462680947028632770999640191867 2\ldots$, $d = 3.28860546617690429718388741565996 5\ldots$, $K = a/(1+bd) - 6.03 \cdot 10^{-23}$ (full 60 digits printed by the script). Displacements on the ray $\{x = 1+\rho,\ y = 1-\rho a/(1+b)\}$: $\delta(0.002) = -9.56 \cdot 10^{-25}$, $\delta(0.006) = +1.89 \cdot 10^{-24}$, $\delta(0.012) = -3.12 \cdot 10^{-23}$, $\delta(0.024) = +2.41 \cdot 10^{-21}$, $\delta(0.045) = -7.01 \cdot 10^{-19}$ (truncated predictions within 0.1–15%); bisection locates the four cycles at $\rho \approx 0.004006,\ 0.008019,\ 0.01614,\ 0.03230$. This is floating-point evidence only (no validated integration).
3. The paper's checkpoints and formulas (§4.4).

9 Trust base, limitations, what a verifier should check first

- **Two independent proofs.** Theorem A rests on the focal-value code + Krawczyk + the bifurcation argument; Theorem C rests only on the validated integrator (~150 lines, `validated_cycles.py`) + Arb + planar topology. Either one alone settles the conjecture. A verifier should check in Theorem C: the a priori enclosure test and the remainder term (§6.4), the crossing bookkeeping (no missed crossing of Σ), and the Jordan-curve/positive-invariance argument.
- **Correctness of the focal-value code.** The certificate checks the polynomials G_k that `focal_exact.py` produces; it does not prove that these are the Lyapunov quantities. Evidence: the derivation in §3–4 (short, checkable by hand), the exact match with the paper's closed-form L_1, the match of an independent generic implementation with every focal value printed in the paper (L_1, L_2, L_3 of three different systems), and an independent method (polar expansion) at the degenerate points, including $u_9(2\pi) = \pi V_4$ at p^*. A verifier should re-derive $\tilde{R} = N_f + iw(cN_I -$

N_f) and, ideally, recompute $L_1, \dots, L_4$ at p^* with an independent CAS/algorithm (e.g. Singular/Maple normal form).

- **Sign lemma (§4.2) and successive perturbations (§6)**: classical; the proof given is self-contained. Check that every perturbed parameter stays on $\mathcal{T}_0$ when Lemma 4.2 is used (it does: Φ maps into $\mathcal{T}_0$), and that the last step only changes K.
- **Interval arithmetic**: Arb (python-flint 0.9.0); comparisons used are the certain ones (< true only if true for all points of the balls). Polynomials are evaluated term by term (no Horner), which only enlarges enclosures.
- **Mass-action interpretation**: exponents are real (generalized mass-action, real complexes), exactly as in the paper's setting (§5: $a > 0$, $b > -1$, $d > 0$, $1 + bd > 0$).
- **Not proved**: exact number (=4) of small cycles and their hyperbolicity (classical, cited in the Remark of §2; for Theorem C, more cycles could in principle hide in the intervals — at least four are proved); the more precise cycle locations of §8.2 (numerical only); anything about large-amplitude cycles.

10 Reproduction

From the project root (Git Bash): `outils/wsl "cd chemistry-open-problems/problems/boros-hofbauer-four-limit-cycles && bash scripts/run_all.sh"`. Individual scripts: `focal_exact.py` (exact $\tilde{V}_k$), `crosscheck.py`, `certify.py`, `uniqueness.py`, `validated_cycles.py` (each prints PASS/FAIL and exits non-zero on failure; seconds each), `polar_check.py` (~5 s), `illustrate_cycles.py` (~3 min, illustration). Full log: `certificates/run_all.log` (35 PASS, 0 FAIL).

11 g_1, g_2, g_3 ($G_k = \kappa_k a^{2k} c^{2k} g_k$, $\kappa = (-\frac{1}{8}, \frac{1}{1152}, -\frac{1}{442368}, \frac{1}{530841600})$)

```
g_1 = 2*a^2*c*s^2 + 2*a^2*c*s + a*c^2*s - 2*a^2*s^2 - 4*a^2*s - a*c*s - 2*a^2 - a*c
+ c^2
```

```
g_2 = 318*a^4*c^3*s^5 + 396*a^4*c^3*s^4 + 159*a^3*c^4*s^4 - 954*a^4*c^2*s^5 + 78*a^4*c^3*s^3
+ 35*a^3*c^4*s^3 - 1998*a^4*c^2*s^4 - 765*a^3*c^3*s^4 + 954*a^4*c*s^5 - 1374*a^4*c^2*s^3
- 1086*a^3*c^3*s^3 - 3*a^2*c^4*s^3 + 3048*a^4*c*s^4 + 1053*a^3*c^2*s^4 - 318*a^4*s^5
- 330*a^4*c^2*s^2 - 409*a^3*c^3*s^2 - 79*a^2*c^4*s^2 + 3804*a^4*c*s^3 + 3391*a^3*c^2*s^3
+ 72*a^2*c^3*s^3 - 1446*a^4*s^4 - 447*a^3*c*s^4 + 2280*a^4*c*s^2 + 3511*a^3*c^2*s^2
+ 900*a^2*c^3*s^2 - 175*a*c^4*s^2 - 2748*a^4*s^3 - 2340*a^3*c*s^3 - 69*a^2*c^2*s^3
+ 570*a^4*c*s + 1173*a^3*c^2*s + 636*a^2*c^3*s + 33*a*c^4*s - 2748*a^4*s^2 - 4134*a^3*c*s^2
- 743*a^2*c^2*s^2 + 175*a*c^3*s^2 - 1446*a^4*s - 3036*a^3*c*s - 1151*a^2*c^2*s +
438*a*c^3*s - c^4*s - 318*a^4 - 795*a^3*c - 477*a^2*c^2 + 159*a*c^3 + 159*c^4
```

`g_3`: 136 terms, see `certificates/G_3.json` (and `g_4`: `certificates/G_4.json`). The JSON format is `{"vars": ["a","c","s"], "terms": [[e_a, e_c, e_s, "p/q"], ...]}` and stores G_k itself (with the constant and the monomial factor).